

I. Project Title

Task and Assignment Scheduling Kit (TASK)

II. Problem Statement

Students often struggle with managing multiple assignments, projects, and daily tasks across different subjects. When deadlines and priorities are tracked across scattered notebooks, mental reminders, or in different group chats, the most critical tasks can easily slip through the gaps. This can lead to last-minute cramming, missed deadlines, increased stress, and poor academic performance. TASK addresses this problem by providing the organized system to log, prioritize, and schedule work.

III. Project Objectives

- Design and implement a core program capable of logging up to 31 tasks with properties for subject, deadline, and priority level.
- Build an automated sorting algorithm that prioritizes logged tasks using a weighted matrix (e.g., deadline urgency) to display a clear "Top Priorities" daily list.
- Develop a notification system logic that flags high-priority assignments due within 24–48 hours to prevent missed deadlines.

IV. Planned Features

1. Allows users to add, edit, or delete tasks with essential fields such as Subject Name, Assignment Title, Due Date/Time, and Priority Rating (High, Medium, Low).
2. Automatically sorts all active tasks using an urgency algorithm to surface the most critical deadlines at the top of the user's view.
3. Provides updates to track each assignment's state (Pending, In Progress, Completed) and archive finished work to declutter the workspace.
4. Generates daily and weekly productivity metrics to give students a clear visual summary of their academic performance.

- Integrates with the user's primary Google account to automatically export task deadlines as events directly into Google Calendar and automatically import assignments and due dates directly from Google Classroom.

V. Planned Inputs and Outputs

Category	Input/Output	Brief Description
Input	Assignment Details	Subject Name, Assignment Title, and Priority Rating (High, Medium, Low)
	Deadline	Target completion date (MM-DD) and specific due time
	Assignment State Updates	User action flags to toggle status (Pending, In Progress, Completed)
	Google Account Auth	User credentials / OAuth token for Google Classroom and Google Calendar access
Output	Priority Dashboard	Sorted list displaying upcoming tasks ordered by urgency and deadline proximity
	Deadline Warnings	High-visibility alerts or flags for tasks due within 24 to 48 hours
	Google Calendar Sync	Automatically created calendar events corresponding to task deadlines
	Summary	Visual performance metrics (e.g., pie chart percentage of tasks completed on time)

VI. Planned Inputs and Outputs

START

```
// STEP 1: LOGIN & GOOGLE SYNC
```

```
DISPLAY "Welcome to TASK!"
```

```
PROMPT user to "Sign in with Google Account"
```

```
IF user signs in successfully THEN
```

```
    DISPLAY "Connected to Google!"
```

```
    FETCH assignments from Google Classroom
```

```
    ADD imported assignments to taskList
```

```
    DISPLAY "Google Classroom assignments synced!"
```

```
ELSE
```

```
    DISPLAY "Continuing in Offline Mode."
```

```
END IF
```

```
// STEP 2: MAIN MENU LOOP
```

```
SET keepRunning = TRUE
```

```
WHILE keepRunning IS TRUE DO
```

```
    DISPLAY "--- MAIN MENU ---"
```

```
    DISPLAY "(1) View Priority Dashboard"
```

```
    DISPLAY "(2) Add a New Task"
```

```
    DISPLAY "(3) Change Task Status"
```

```
    DISPLAY "(4) View Progress Analytics"
```

```
    DISPLAY "(5) Exit"
```

```
INPUT choice
```

```
// --- CHOICE 1: VIEW DASHBOARD ---
```

```
IF choice == 1 THEN
```

```
    IF taskList IS EMPTY THEN
```

```
        DISPLAY "No tasks found!"
```

```
    ELSE
```

```
        SORT taskList BY dueDate (soonest first) AND priority (High to Low)
```

```
        DISPLAY "=== YOUR TASK DASHBOARD ==="
```

```
        FOR EACH task IN taskList DO
```

```
        IF task.status IS NOT "Completed" THEN
            DISPLAY task.title + " | " + task.subject + " | Due: " + task.dueDate + "
| Priority: " + task.priority
```

```
        // Check for upcoming deadline alert
        IF task.dueDate IS within 2 days THEN
            DISPLAY "! WARNING: Due very soon !"
        END IF
    END IF
END FOR
END IF
```

```
// --- CHOICE 2: ADD A TASK ---
```

```
ELSE IF choice == 2 THEN
```

```
    INPUT title
    INPUT subject
    INPUT dueDate
    INPUT priority // High, Medium, or Low
```

```
    SET status = "Pending"
    SAVE task (title, subject, dueDate, priority, status) TO taskList
    DISPLAY "Task successfully saved!"
```

```
    IF connected to Google THEN
        EXPORT dueDate TO Google Calendar
        DISPLAY "Added to your Google Calendar!"
    END IF
```

```
// --- CHOICE 3: CHANGE TASK STATUS ---
```

```
ELSE IF choice == 3 THEN
```

```
    DISPLAY "Enter the name of the task to update:"
    INPUT targetTask
```

```
    FIND targetTask IN taskList
```

```
    IF task found THEN
```

```
        DISPLAY "Choose status: 1. Pending 2. In Progress 3. Completed"
        INPUT newStatus
```

```

        UPDATE task status TO newStatus
        DISPLAY "Status updated!"
    ELSE
        DISPLAY "Task not found."
    END IF

// --- CHOICE 4: VIEW ANALYTICS ---
ELSE IF choice == 4 THEN
    SET total = COUNT all tasks IN taskList
    SET completed = COUNT tasks WHERE status == "Completed"

    IF total > 0 THEN
        SET percentage = (completed / total) * 100
        DISPLAY "=== YOUR PROGRESS ==="
        DISPLAY "Total Tasks: " + total
        DISPLAY "Completed Tasks: " + completed
        DISPLAY "Completion Rate: " + percentage + "%"
    ELSE
        DISPLAY "No tasks logged yet."
    END IF

// --- CHOICE 5: EXIT ---
ELSE IF choice == 5 THEN
    SET keepRunning = FALSE
    DISPLAY "Goodbye! Stay productive!"

ELSE
    DISPLAY "Invalid option. Please choose 1 to 5."
END IF

END WHILE

END

```